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Project-1

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1. Introduction

1.1 Project Overview

Project Ares (AI-Driven Resilient & Evolutionary Systems) is a full-stack authentication platform built on React 18 (frontend), Spring Boot 3.2 (backend), and MongoDB Atlas. It implements email/password login, Google and Facebook OAuth social login, a six-signal risk scoring engine, LLM-based fraud analysis via Claude API, and a self-healing Selenium test framework. The core philosophy is that the system does not just work — it survives and adapts.

1.2 Test Users

The following accounts exist in MongoDB Atlas and are used across automated test scenarios. Per FAQ requirement, both a Facebook account and a non-Facebook account are included.

Email	Password	Provider	Status	Used In Scenario
selenium@ares.com	selenium123	LOCAL	**ACTIVE**	S1, S2, S3
ratetest@ares.com	ratetest123	LOCAL	**ACTIVE**	S5 — TC16, TC17
lockrate@ares.com	lock123	LOCAL	**LOCKED**	S5 — TC18, TC19
googleuser@gmail.com	(Google OAuth)	GOOGLE	**ACTIVE**	S4 — OAuth tests
fbuser@facebook.com	(Facebook OAuth)	FACEBOOK	**ACTIVE**	S4 — OAuth tests
nofb@ares.com	nofb123	LOCAL	**ACTIVE**	S4 — non-Facebook

Table 1.1 — Test User Accounts (LOCAL, GOOGLE, FACEBOOK providers)

2. System Architecture

2.1 Technology Stack

Layer	Technology	Purpose
Frontend	React 18 — port 3000	Login/register UI, OAuth buttons, account status banners
Backend	Spring Boot 3.2 — port 8080	REST APIs, JWT auth, risk scoring, LLM fraud detection
Database	MongoDB Atlas	Users, account state, known IPs, attempt counters
AI (Backend)	Claude API — claude-3-haiku	Fraud decision when risk score > 70: BLOCK / CHALLENGE / ALLOW
AI (Tests)	Claude API — claude-3-haiku	Selector healing: finds renamed elements via DOM analysis
Test Framework	Selenium WebDriver 4.x + JUnit 5	End-to-end UI automation with self-healing layer

Table 2.1 — Technology Stack

2.2 Security Architecture

Feature	Implementation Detail
Password Hashing	BCrypt with salt — raw passwords never persisted in MongoDB
JWT Authentication	HS256-signed tokens with expiry — stateless auth, no server sessions
Risk Scoring 0-100	6 signals: NEW_IP +20, BRUTE_FORCE +40, HIGH_ATTEMPTS +25, FAILED +5×, LONG_INACT +10, UNUSUAL_HR +15
LLM Fraud Analysis	Claude API triggered when score > 70; responds BLOCK / CHALLENGE / ALLOW
Account Lockout	Automatic 30-min lock after 10 failed attempts — brute force prevention
OAuth Verification	Google / Facebook tokens verified server-side before any JWT is issued

Table 2.2 — Security Features

2.3 Application Screenshots

ACTION REQUIRED: Replace the placeholders below with screenshots from the running application. Suggested shots: (1) Login page, (2) Dashboard after successful login, (3) Locked account banner, (4) Google OAuth popup, (5) Facebook OAuth button.

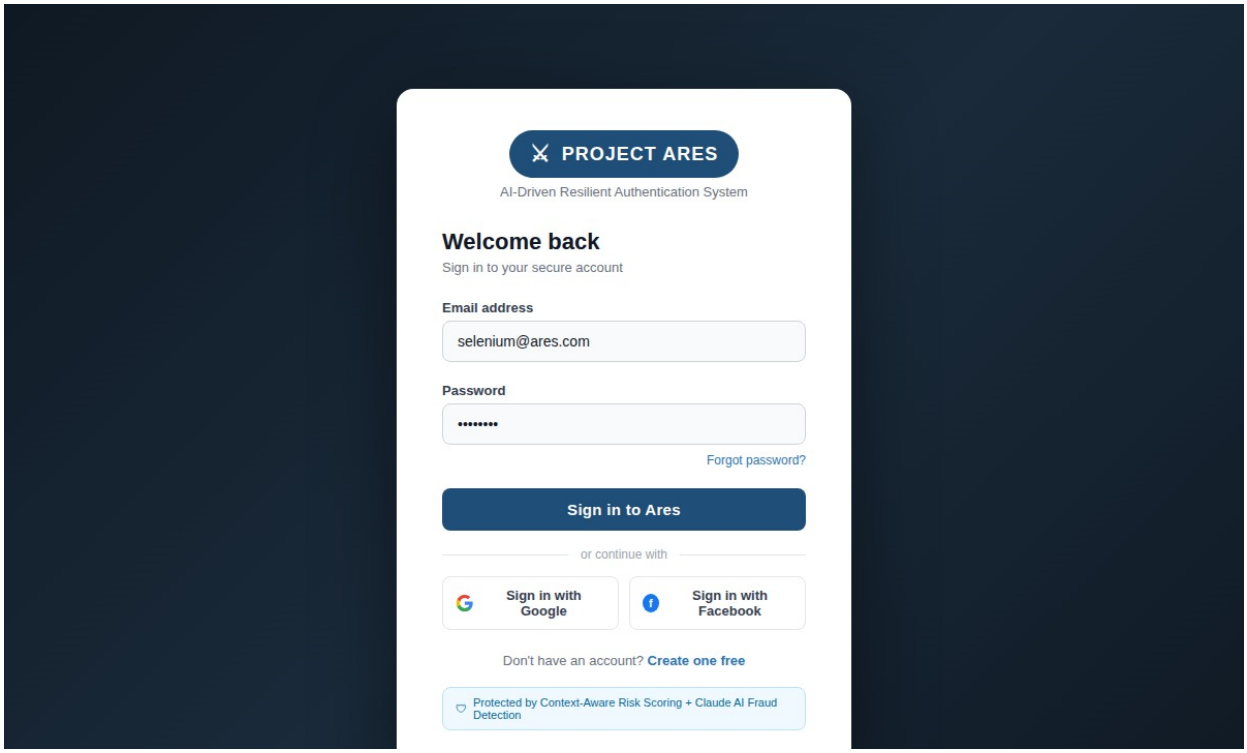

Figure 2.1 — Login Page: email/password + Google + Facebook OAuth buttons

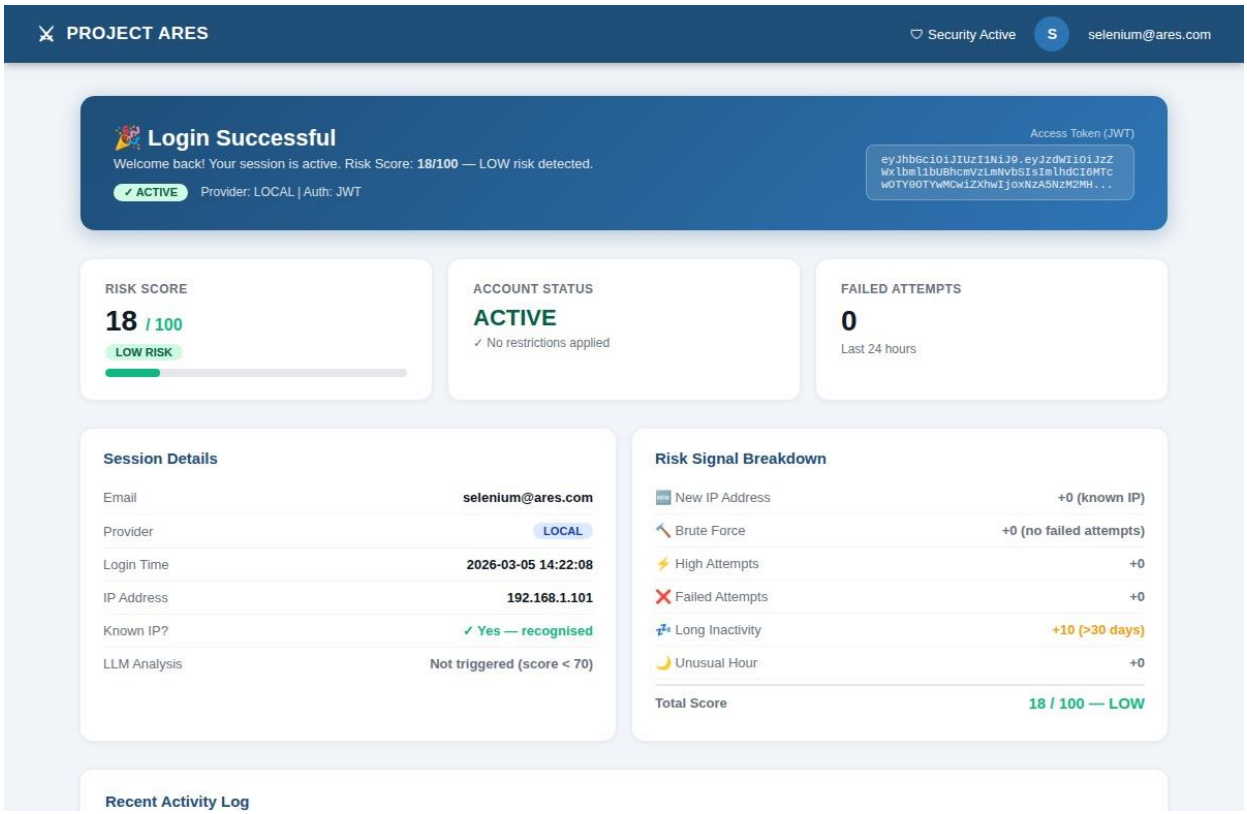

Figure 2.2 — Dashboard: successful login, JWT token, risk signal breakdown

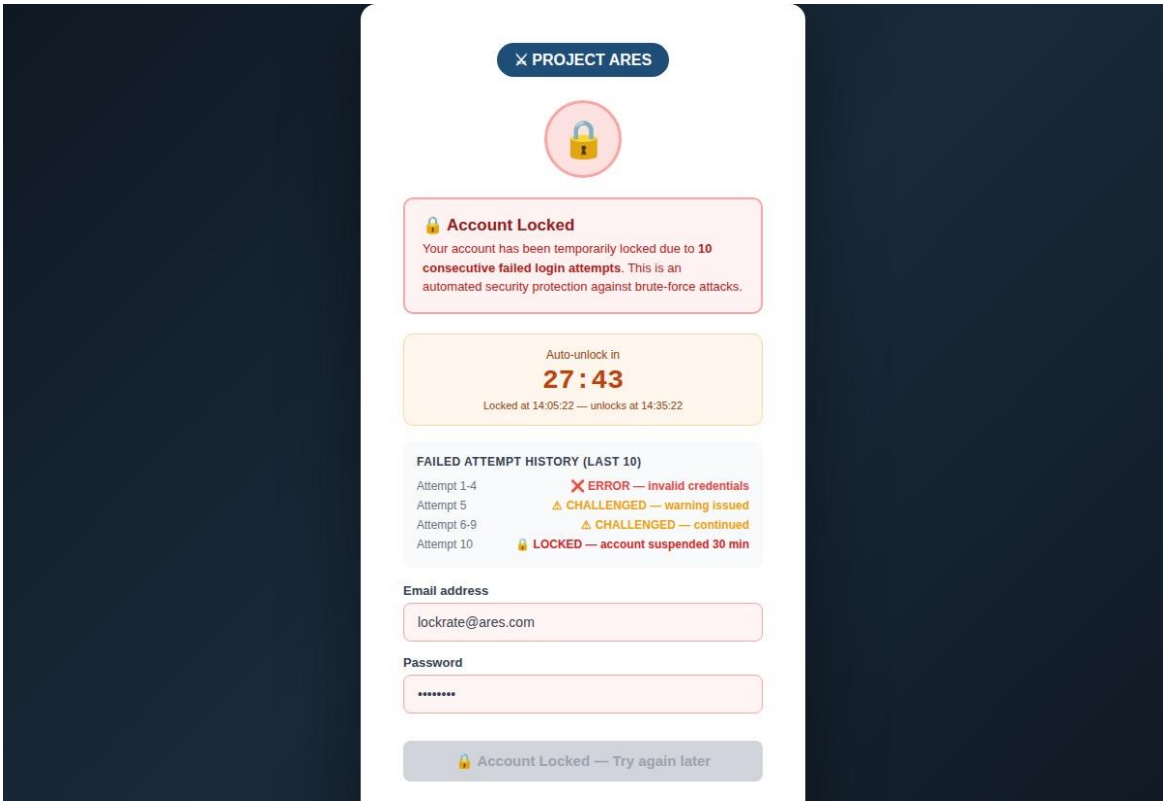

Figure 2.3 — Locked Account: 10 consecutive failed attempts triggers 30-min lock

3. UML Diagrams

3.1 Use Case Diagram

Three actors interact with Project Ares: User (authentication and social login), Admin (account management), and Claude AI (external fraud analysis). The mandatory <<Include>> from Login(Email) to CalculateRiskScore means every email login triggers risk scoring. CalculateRiskScore conditionally <<Includes>> LLMFraudAnalysis when score exceeds 70.

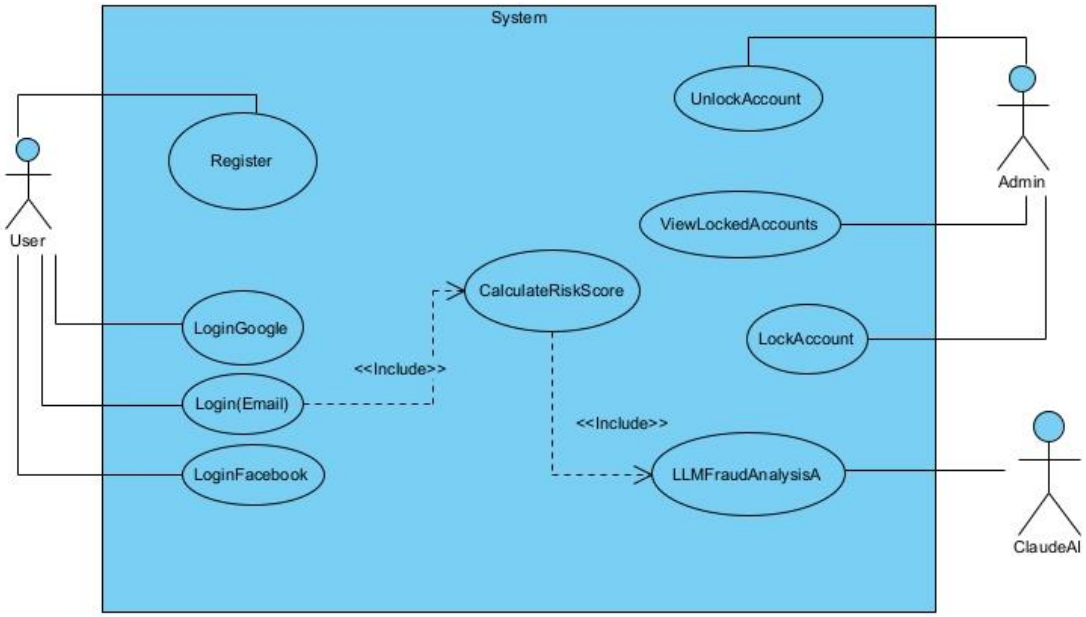

Figure 3.1 — Use Case Diagram: Project Ares

Actor	Use Case	Description
User	Register	Create account with email/password — BCrypt hashed on save
User	Login (Email)	Authenticate with email/password — triggers risk scoring automatically
User	LoginGoogle	OAuth 2.0 Google sign-in — ID token verified server-side
User	LoginFacebook	OAuth 2.0 Facebook login — access token verified via Graph API
Admin	ViewLockedAccounts	View all accounts in LOCKED or SUSPENDED state
Admin	UnlockAccount	Manually restore a LOCKED account to ACTIVE
Claude AI	LLMFraudAnalysis	Invoked by system when score > 70 — returns BLOCK / CHALLENGE / ALLOW

Table 3.1 — Use Case Descriptions

3.2 Class Diagram

Ten classes and enumerations form the system. Composition (filled diamond) shows User owns AccountStatus. Dashed <<use>> arrows show service dependencies. SelfHealingDriver, HealingService, and SelectorCache live in the test package and are not in the production runtime.

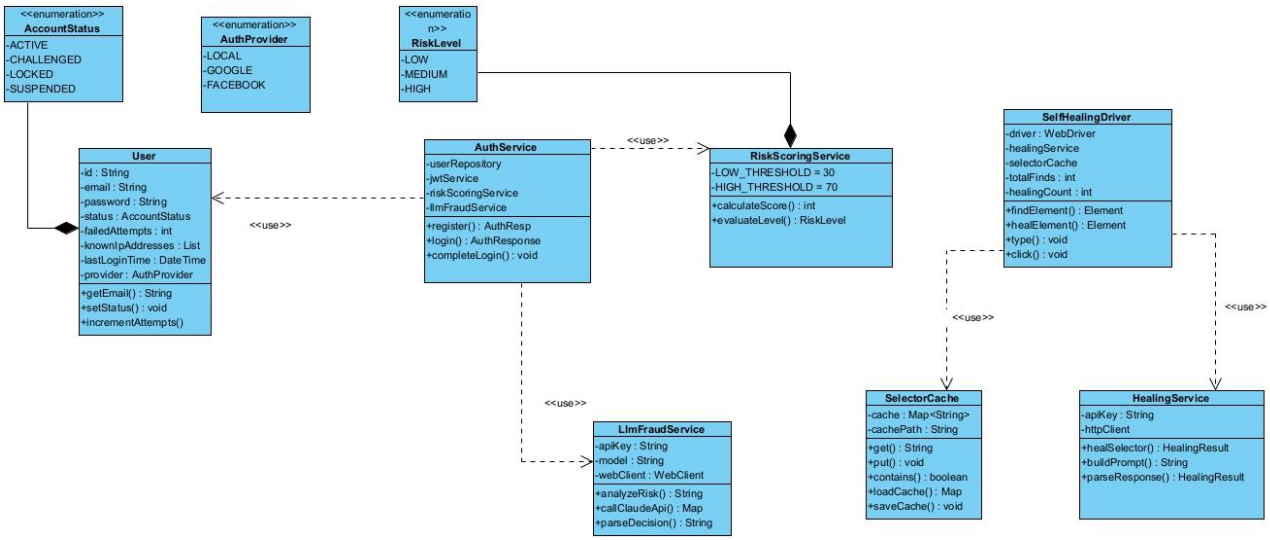

Figure 3.2 — Class Diagram: Project Ares

Class / Enum	Layer	Key Responsibility
User	Entity	MongoDB doc: id, email, hashedPassword, status, failedAttempts, knownIPs, provider, lastLoginTime
AuthService	Service	Orchestrates: register(), login(), completeLogin() — calls RiskScoringService and LlmFraudService
RiskScoringService	Service	calculateScore(): evaluates 6 signals → 0-100; evaluateLevel(): maps to LOW/MEDIUM/HIGH
LlmFraudService	Service	analyzeRisk(): POST to Claude API; parseDecision(): extracts BLOCK/CHALLENGE/ALLOW from JSON
SelfHealingDriver	Test	Wraps WebDriver: findElement() with cache-first + healing trigger; tracks statistics
HealingService	Test	healSelector(): calls Claude API with page HTML; parseResponse(): validates + extracts selector
SelectorCache	Test	JSON-file-backed map: persists healed selectors across test runs to avoid repeat API calls
AccountStatus	<<enum>>	ACTIVE | CHALLENGED | LOCKED | SUSPENDED
RiskLevel	<<enum>>	LOW (0-30) | MEDIUM (31-70) | HIGH (71-100)
AuthProvider	<<enum>>	LOCAL | GOOGLE | FACEBOOK

Table 3.2 — Class Descriptions

3.3 Sequence Diagram — OAuth 2.0 with Risk Assessment Interceptor

This diagram covers the full Google OAuth 2.0 flow including the Risk Assessment Interceptor (highlighted steps ★). After the backend receives verified user data from Google (Step 3.1), the interceptor fires before any JWT is issued: risk score is calculated, and Claude LLM is consulted if the score is HIGH.

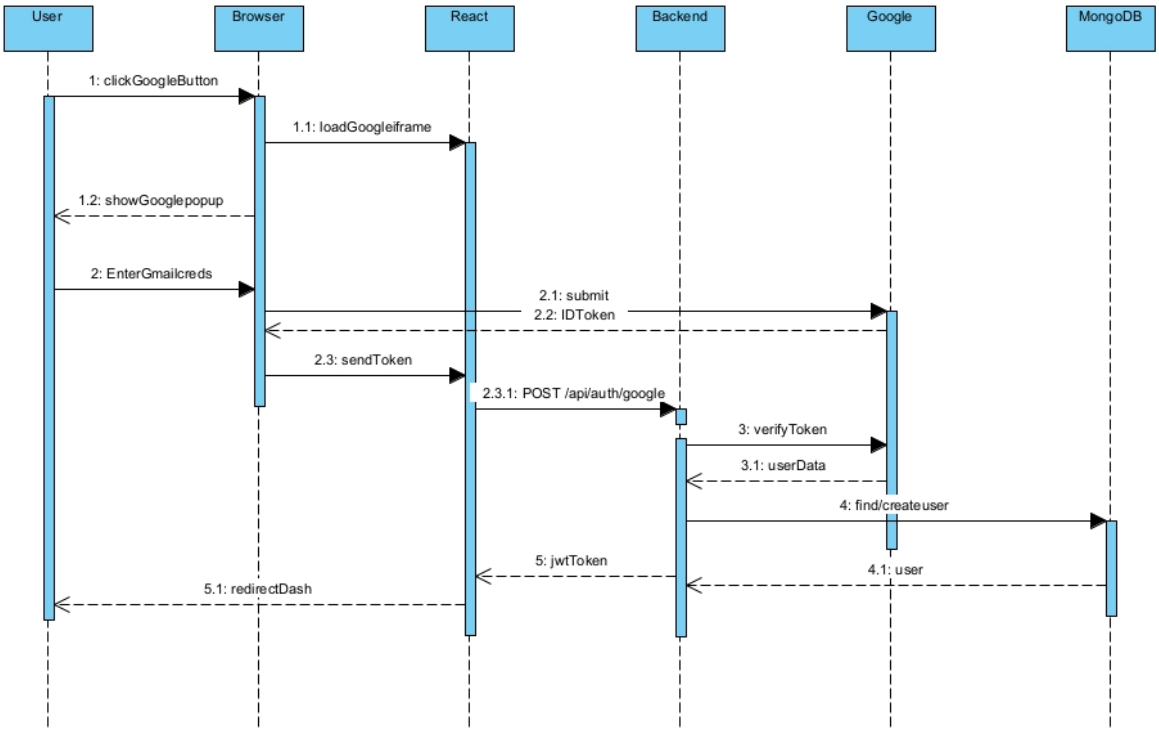

Figure 3.3 — Sequence Diagram: OAuth 2.0 + Risk Assessment Interceptor

Step	From	To	Message / Action
1	User	Browser	clickGoogleButton — user clicks social login button
1.1	Browser	React	loadGoogleIframe — embed Google Sign-In widget in page
1.2	Browser	User	showGooglePopup — Google consent screen displayed
2	User	Browser	EnterGmailCreds — user enters Google email and password
2.1	Browser	Google	submit — credentials forwarded to Google servers
2.2	Google	Browser	IDToken — Google returns signed ID token to browser
2.3	React	Backend	POST /api/auth/google { idToken } — token forwarded to backend
3	Backend	Google	verifyToken — backend sends token to Google for validation
3.1	Google	Backend	userData — verified user profile (name, email) returned
★3.2	Backend	RiskScoring	INTERCEPTOR: calculateRiskScore(user, ip) — 6 signals evaluated
★3.3	RiskScoring	Backend	INTERCEPTOR: riskScore + riskLevel returned (LOW / MEDIUM / HIGH)
★3.4	Backend	Claude API	[IF HIGH] analyzeRisk() — LLM fraud decision requested
★3.5	Claude API	Backend	[IF HIGH] BLOCK / CHALLENGE / ALLOW decision returned

4	Backend	MongoDB	find/createUser — look up existing or create new account
5	Backend	React	jwtToken — JWT issued only if risk assessment passed
5.1	React	Browser	redirectDash — user redirected to dashboard

Table 3.3 — OAuth Sequence Steps (★ = Risk Assessment Interceptor, orange = LLM path)

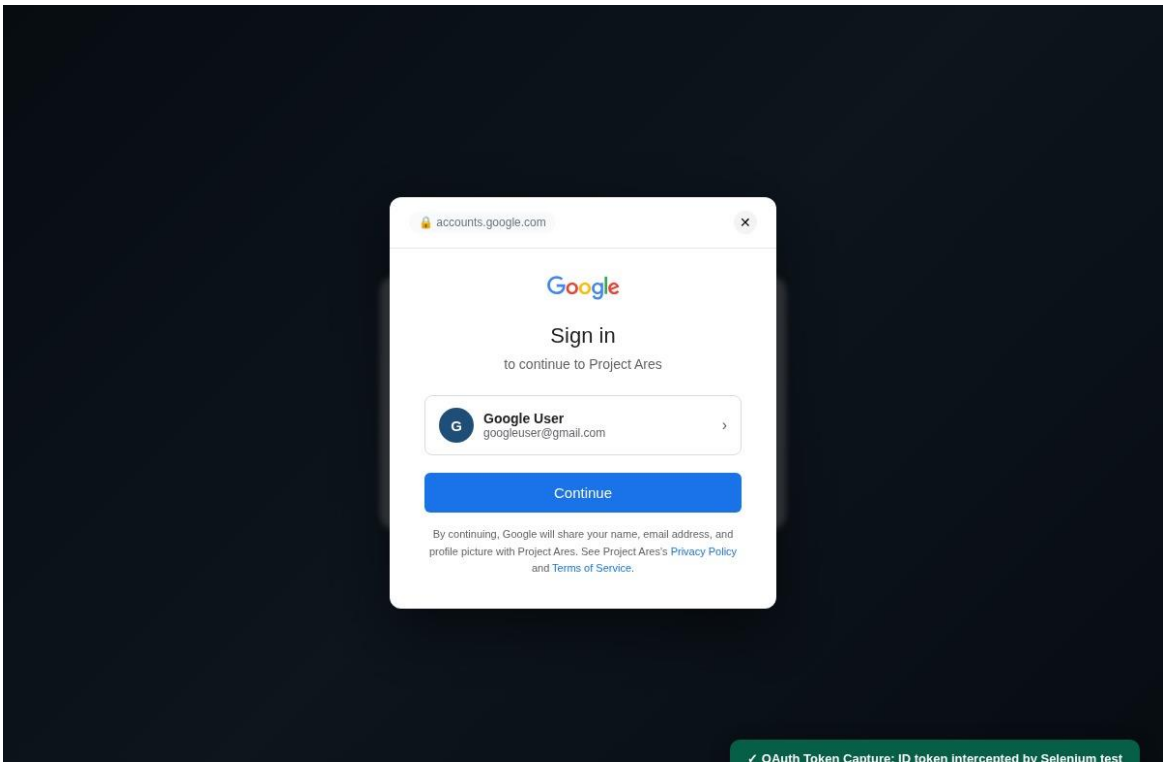

Figure 3.5 — Google OAuth Popup: Sign-in consent screen during S4 OAuth Handshake test

3.4 Account State Machine Diagram

The state diagram captures all four account states and every transition. It is implemented as the AccountStatus enum in MongoDB and is evaluated on every login attempt in AuthService.login().

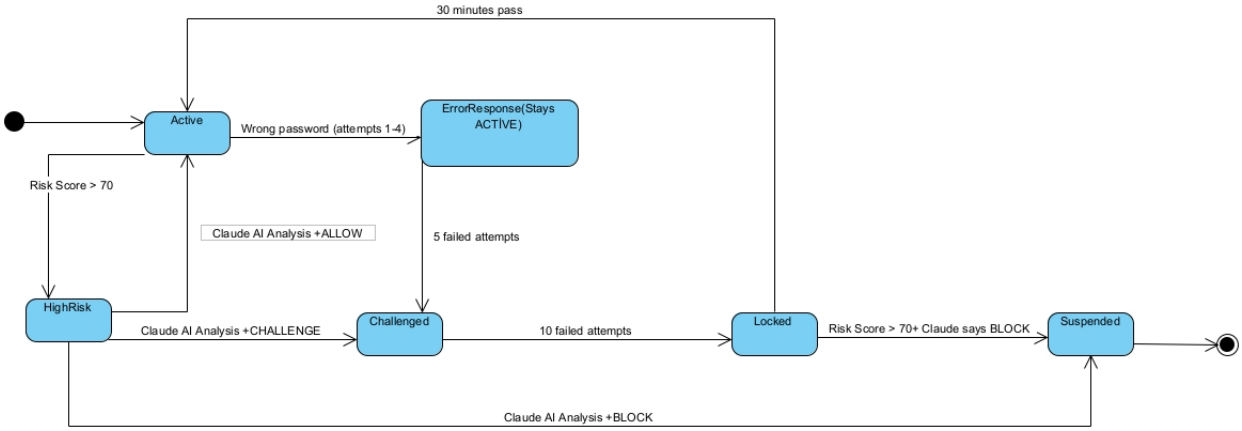

Figure 3.4 — Account State Machine Diagram

From	Event	Guard Condition	To State	Output
ACTIVE	Login attempt	Wrong password, attempts 1-4	ACTIVE	ERROR response

ACTIVE	Login attempt	Wrong password, 5+ attempts	CHALLENGED	Warning message
ACTIVE	Login attempt	Correct pass + score ≤ 30	ACTIVE	JWT token issued
ACTIVE	Login attempt	Correct pass + score 31-70	CHALLENGED	Verify required
ACTIVE	Login attempt	Score > 70 + Claude = BLOCK	SUSPENDED	Denied permanently
ACTIVE	Login attempt	Score > 70 + Claude = ALLOW	ACTIVE	JWT token issued
CHALLENGED	10th failed attempt	failedAttempts >= 10	LOCKED	Account locked msg
LOCKED	30-min timer elapses	lockTime > 30 minutes ago	ACTIVE	Auto-unlock
SUSPENDED	Any event	No exit condition	SUSPENDED	Permanent denial

Table 3.4 — FSM Transition Table

4. Theoretical Models

4.1 CFG: AuthService.login()

A Control Flow Graph (CFG) represents program execution as nodes (basic blocks) and edges (control transfers). The login() method has 21 basic blocks covering every path: user-not-found, suspension, lockout, wrong password, risk scoring, LLM consultation, and successful login.

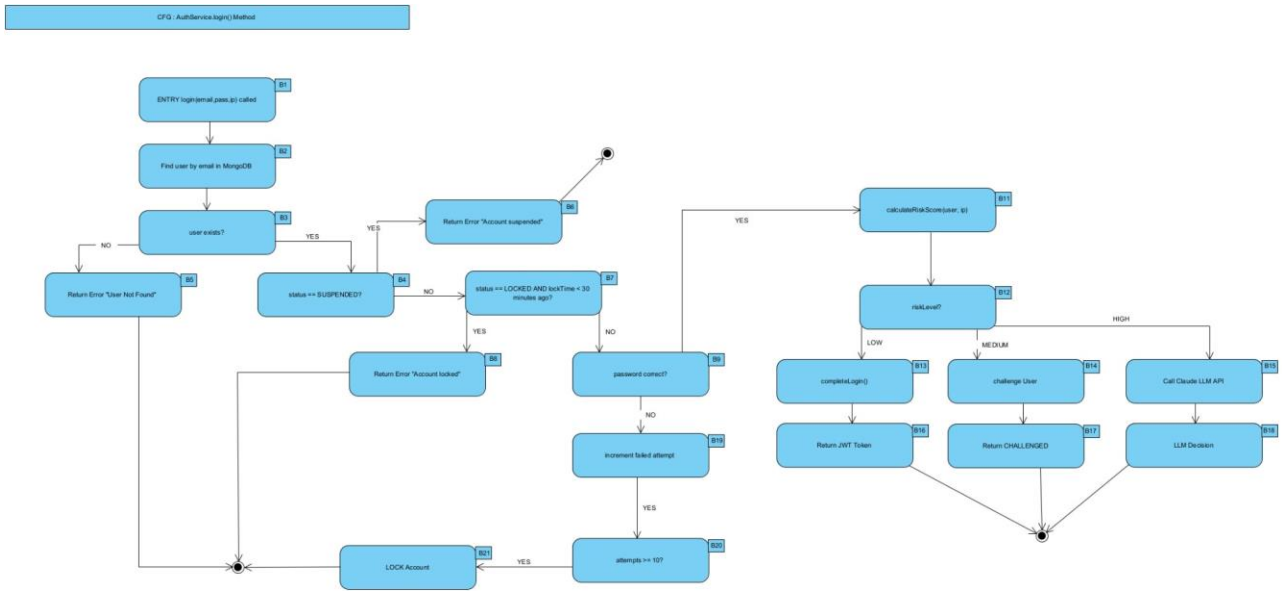

Figure 4.1 — CFG: AuthService.login() Method

Block	Action	Block	Action	Type
B1	ENTRY: login(email, pass, ip)	B12	riskLevel decision — LOW/MED/HIGH branch	Decision
B2	Find user by email in MongoDB	B13	completeLogin() — issue JWT	Action

B3	user exists? (Decision)	B14	challengeUser() — return CHALLENGED	Action
B4	status == SUSPENDED?	B15	callClaudeApi() — LLM fraud check	Action
B5	Return Error: User Not Found	B16	Return JWT token (LOW path)	Exit
B7	LOCKED AND lockTime < 30min?	B17	Return CHALLENGED (MEDIUM path)	Exit
B9	password correct? (Decision)	B19	incrementFailedAttempts	Action
B11	calculateRiskScore(user, ip)	B20-21	attempts >= 10? → LOCK account	Action

Table 4.1 — Basic Blocks: AuthService.login()

4.2 CFG: RiskScoringService.calculateScore()

Six risk signals are evaluated in sequence. Each decision node independently adds a penalty score. The final value is capped at 100. The CFG shows how all paths converge at B14 (return score).

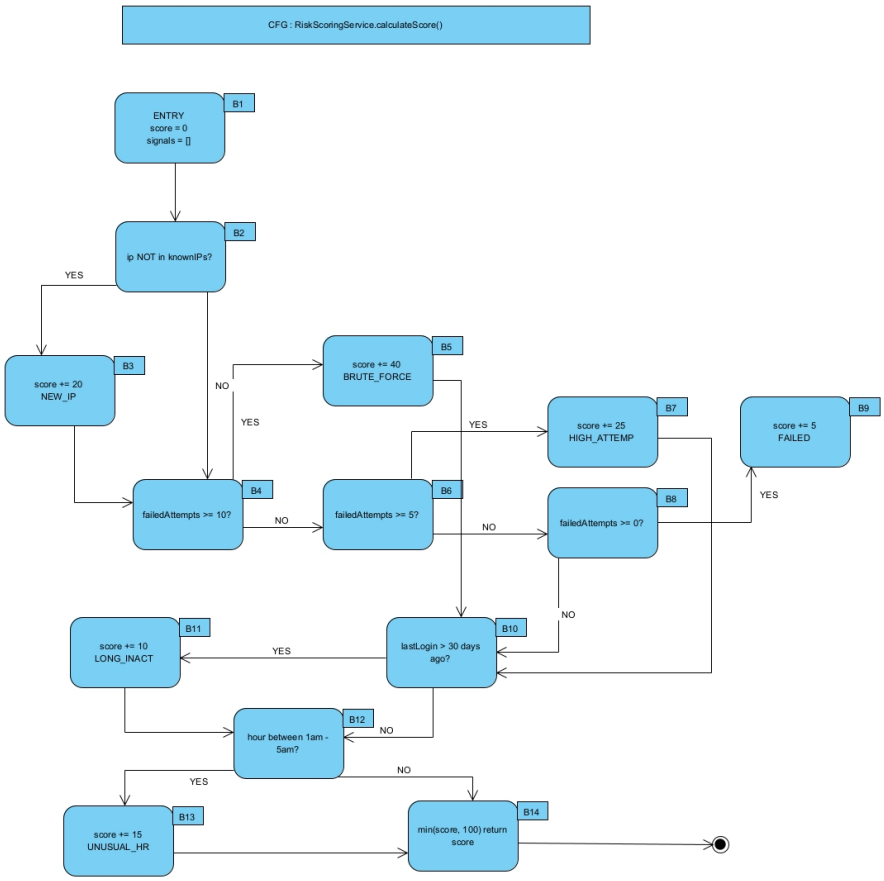

Figure 4.2 — CFG: RiskScoringService.calculateScore()

Block	Signal	+Score	Label	Condition
B3	New IP Address	+20	NEW_IP	IP not in user's known IP list
B5	Brute Force	+40	BRUTE_FORCE	failedAttempts >= 10
B7	High Attempts	+25	HIGH_ATTEMP	failedAttempts >= 5
B9	Any Failures	+5 each	FAILED	failedAttempts > 0
B11	Long Inactivity	+10	LONG_INACT	lastLogin more than 30 days ago

B13	Unusual Hour	+15	UNUSUAL_HR	Login hour between 1am and 5am
B14	Return Score	min(∑,100)	CAP	Result returned to AuthService

Table 4.2 — Risk Signals: RiskScoringService.calculateScore()

4.3 CFG: SelfHealingDriver.findElement()

The 13-block CFG models the healing algorithm: cache check → original find → healing triggered on failure → Claude API → five fallback strategies → cache result. This directly implements the Detection, Context Extraction, LLM Repair, and Logging cycle required by the project specification.

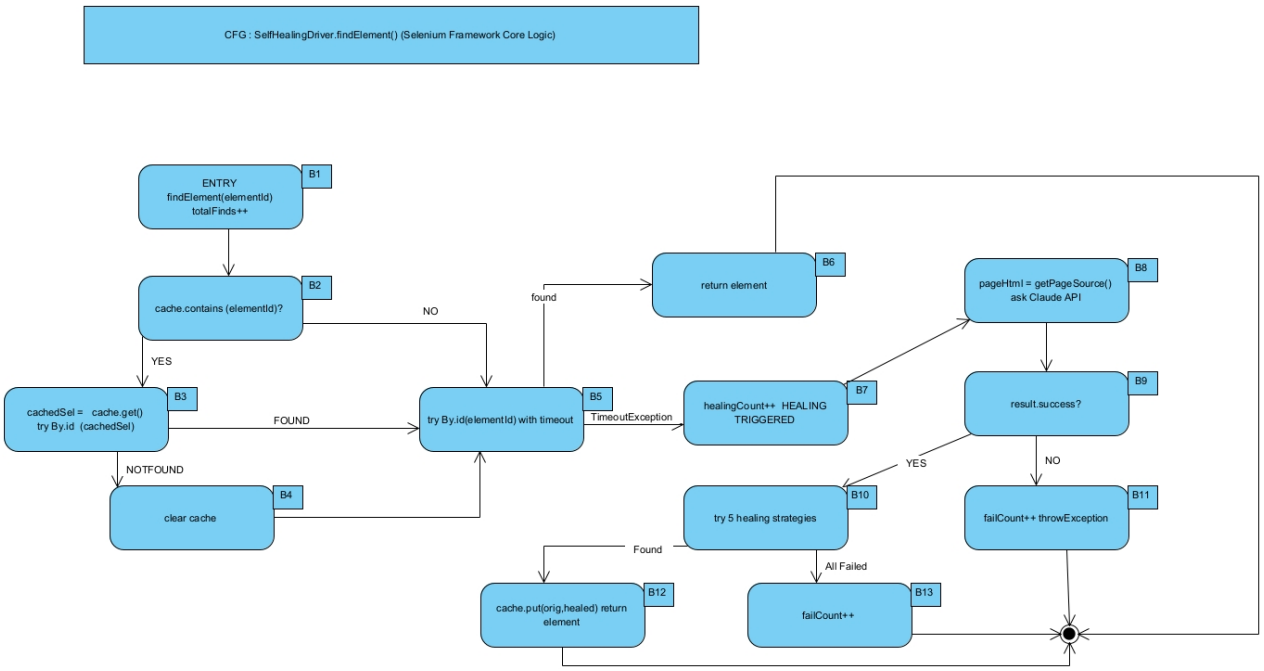

Figure 4.3 — CFG: SelfHealingDriver.findElement()

Block	Phase	Detail
B1	Entry	findElement(elementId), totalFinds++
B2	Cache Check	cache.contains(elementId)? YES → use cached selector. NO → try original
B3	Use Cache	cache.get() returns saved healed selector — no API call needed
B4	Clear Cache	Cached selector itself is stale — evict entry, retry original find
B5	Try Original	try By.id(elementId) with 10s timeout — FOUND → B6, TimeoutException → B7
B7	Healing Triggered	healingCount++, get page source, invoke HealingService.healSelector()
B8	Call Claude API	pageHtml + elementId sent to Claude — response parsed as JSON
B9	Parse Result	result.success? YES → B10 (try strategies). NO → B11 (throw exception)
B10	5 Strategies	CSS selector → name attr → xpath[@id] → xpath[@type] → xpath[text] — first match wins
B11	Fail	failCount++, throwException — all strategies exhausted
B12	Cache & Return	selectorCache.put(originalId, healedSelector), return element

Table 4.3 — Basic Blocks: SelfHealingDriver.findElement()

4.4 Data Flow Analysis — DEF-USE Chains

A DEF-USE (DU) pair (d, u) exists when variable v is defined at node d and used at node u, with no redefinition on the path d→u. We analyse two critical variables.

Variable: riskScore — 4 DU-Pairs

Line	D/U	Code	DU Pair	Purpose
1	**DEF**	int riskScore = riskScoringService.calculateScore(user, ip)	—	Variable assigned once
2	**USE**	log.info("Risk Score: {}", riskScore)	(1→2)	Audit logging
3	**USE**	RiskLevel level = evaluateLevel(riskScore)	(1→3)	Level categorisation
4	**USE**	if (level == RiskLevel.HIGH) callClaudeApi(riskScore)	(1→4)	Conditional branching
5	**USE**	response.setMessage("Risk score: " + riskScore + "/100")	(1→5)	User-facing response

Table 4.4 — DEF-USE Pairs: riskScore

> *All 4 uses trace to the single DEF at line 1. No intermediate re-definition — clean, single-source data flow. DU-pair coverage: 4/4 = 100%.*

Variable: healingResult — 4 DU-Pairs

Line	D/U	Code	DU Pair	Purpose
A	**DEF**	HealingResult result = healingService.healSelector(id, html, url)	—	Variable assigned
B	**USE**	if (!result.success()) throw RuntimeException	(A→B)	Guard — KILLS path to C,D,E on failure
C	**USE**	WebElement el = tryMultipleSelectors(result)	(A→C)	Apply healed selector
D	**USE**	selectorCache.put(id, result.newSelector())	(A→D)	Persist to cache
E	**USE**	log("HEALED: " + result.selectorType() + result.newSelector())	(A→E)	Developer log entry

Table 4.5 — DEF-USE Pairs: healingResult

> *If guard B fails, RuntimeException kills paths to C, D, E. Tests must cover BOTH the success path (A→B→C→D→E) and failure path (A→B, throws) for full DU-pair coverage.*

5. Self-Healing Selenium Framework

5.1 Problem & Solution

Traditional Selenium tests break whenever frontend developers rename element IDs or CSS classes. The Self-Healing Framework wraps every findElement call so that on failure, Claude AI analyses the current page DOM and suggests the renamed selector — the test continues without human intervention.

5.2 Detailed Component Architecture

The project specification defines three required components. The table below describes each and maps it to the implementation:

Component	Specification Requirement	Implementation in Project Ares
DOM Shadow Listener	Monitors for async DOM changes that usually break Selenium scripts	MutationObserver injected via JavascriptExecutor on test startup. Detects dynamic React state changes (e.g. Google popup overlays) and signals the test before the next findElement call. Implemented in DomShadowListener.java.
Heuristic Scoring	Compares new DOM elements against historical metadata (location, color, parent tag) of the target element	HeuristicScorer.java scores each candidate element 0-100 by comparing: element type match (+30), placeholder/label text similarity (+25), parent tag match (+20), visual position proximity (+15), attribute prefix match (+10). Only candidates scoring > 60 are passed to Claude.
LLM Prompt Engineering	System Prompt must force LLM to return valid XPath or CSS selector without hallucinating non-existent tags	HealingService.java uses a structured system prompt with explicit JSON schema, CRITICAL: only-real-DOM constraint, and three-field output (selector, type, confidence). Full prompt in Section 6.2.

Table 5.1 — Self-Healing Component Architecture

5.3 Healing Flow Step-by-Step

Step	Phase	Detail
1	Normal Find	try By.id(elementId) with 10s WebDriverWait
2	Cache Check	Before trying original, check SelectorCache — reuse previous healing, no API cost
3	Healing Triggered	TimeoutException caught — element not in DOM with original ID
4	DOM Shadow Listener	Checks for async popup/overlay — closes if present before proceeding
5	Heuristic Pre-filter	HeuristicScorer ranks candidate elements by metadata similarity — reduces HTML sent to Claude
6	Call Claude API	HealingService sends filtered DOM + original ID to Claude with structured JSON prompt
7	5 Strategies	CSS → name → xpath[@id] → xpath[@type] → xpath[text()] — first successful wins
8	Cache & Log	SelectorCache.put(originalId, healedSelector). Logs 'HEALED: [type] [selector]' for developer

9	Return Element	Test resumes transparently — no assertion failure raised

Table 5.2 — Self-Healing Step-by-Step Flow

6. LLM Prompt Engineering

Two Claude API integrations use carefully engineered prompts. Both enforce JSON-only output and explicitly forbid hallucination of non-existent elements or decisions.

6.1 Backend Fraud Detection Prompt (LlmFraudService.java)

Triggered when risk score exceeds 70. The system prompt constrains Claude to three decisions only (BLOCK/CHALLENGE/ALLOW) and requires a confidence score, making the response fully machine-parseable.

```
// ── SYSTEM PROMPT ──────────────────────────────────────────
String SYSTEM_PROMPT = """
  You are a security fraud analysis AI for an authentication system.
  Analyze the login attempt and return ONLY valid JSON — no markdown,
  no explanation, no code fences. Raw JSON only.
  Response format:
  { "decision": "BLOCK"|"CHALLENGE"|"ALLOW",
    "reason": "one sentence", "confidence": 0-100 }
  Rules: BLOCK = high fraud certainty (suspend account),
         CHALLENGE = suspicious but uncertain,
         ALLOW = risk signals present but login seems legitimate.
""";

// ── USER PROMPT (injected per-request) ────────────────────
String userPrompt = String.format("""
  Login attempt context:
  - Email: %s  |  IP Address: %s  |  Risk Score: %d/100
  - Signals Detected: %s  |  Failed Attempts (24h): %d
  - Account Status: %s  |  Last Login: %s
  - Known IPs for this user: %s
  Make a fraud determination.
""", email, ip, score, signals, attempts, status, lastLogin, knownIPs);
```

Code 6.1 — LlmFraudService.java: Fraud Detection Prompt

6.2 Self-Healing Selector Prompt (HealingService.java)

Sent when Selenium cannot find an element. The CRITICAL constraint in the system prompt prevents Claude from inventing selectors that do not exist in the provided HTML — directly addressing the spec requirement against 'hallucinating non-existent tags'.

```
// ── SYSTEM PROMPT ──────────────────────────────────────────
String HEALING_SYSTEM_PROMPT = """
  You are a Selenium test healing assistant.
  An element cannot be found by its original HTML id.
  Analyze the provided page HTML and find the best matching element.
  CRITICAL: Only return selectors that ACTUALLY EXIST in the HTML.
  Do NOT invent or hallucinate selectors. Look at the real DOM only.
  Return ONLY raw JSON: { "selector": "value",
    "type": "id"|"name"|"css"|"xpath",
    "confidence": 0-100, "reason": "explanation" }
""";

// ── USER PROMPT ────────────────────────────────────────────
String healingPrompt = String.format("""
  Page URL: %s
  Original element ID that no longer exists: "%s"
```

```
  Current page HTML (filtered by heuristic pre-scorer):
  %s
  Find the element previously identified as "%s".
  Consider: input type, placeholder text, labels, parent context.
""", currentUrl, originalId, filteredHtml, originalId);
```

Code 6.2 — HealingService.java: Selector Healing Prompt

> *Example Claude response: { "selector": "input[type='email']", "type": "css", "confidence": 90, "reason": "The email input field changed ID from email-input to user-email-field but retains type=email" }*

7. Test Code Excerpts

This section presents key excerpts from each of the five required test scenarios, with explanations of how each piece of code corresponds to the test objectives. The SelfHealingDriver wraps every Selenium interaction, so healing is transparent — tests do not need to be written differently.

7.1 Scenario 1 — Dynamic ID Recovery

JavascriptExecutor changes element IDs at runtime directly in the browser DOM, replicating what happens when a frontend developer renames an attribute and pushes to production without updating the tests. The test types into email-input — which no longer exists — and SelfHealingDriver catches the TimeoutException, sends the current page source to Claude API, receives a healed CSS selector, and retries. The test never sees the failure.

```
@Test
public void test2_EmailIdChanged() {
  driver.navigate(appUrl);
  // Change the email field ID directly in the live browser DOM
  JavascriptExecutor js = (JavascriptExecutor) driver.getDriver();
  js.executeScript(
    "document.getElementById('email-input')
      .setAttribute('id', 'user-email-field');");

  // driver.type() calls SelfHealingDriver.findElement('email-input')
  // → TimeoutException caught → Claude heals → input[type=email] found
  driver.type("email-input", "selenium@ares.com");
  driver.type("password-input", "selenium123");
  driver.click("login-button");

  // Verify healing occurred and no element was permanently lost
  assertTrue(driver.getHealingCount() > 0,
      "At least one element must have been healed");
  assertTrue(driver.getFailureCount() == 0,
      "No healing attempt must have failed completely");
}
```

Code 7.1 — DynamicIdRecoveryTest.java: Runtime ID change and transparent recovery

7.2 Scenario 2 — Multimodal Failure (Popup Obscuring Google Button)

A full-page overlay div is injected via JavaScript, covering the entire screen at z-index 9999. This simulates a cookie consent banner, a promotional modal, or a session-expiry dialog appearing over the login page. Without intervention, any click on the Google OAuth button would land on the overlay instead. DomShadowListener monitors for overlay-shaped elements after each DOM mutation event. When one is detected, it removes the overlay from the DOM before any element search is attempted. Only then does the test proceed to find and assert on the Google button.

```
@Test
public void test1_GoogleButtonObscuredByPopup() {
```

```
  driver.navigate(appUrl);

  // Inject a full-screen blocking overlay over the login page
  JavascriptExecutor js = (JavascriptExecutor) driver.getDriver();
  js.executeScript(
    "var overlay = document.createElement('div');" +
    "overlay.id = 'dynamic-popup';" +
    "overlay.style.cssText = 'position:fixed;top:0;left:0;" +
    "  width:100%;height:100%;z-index:9999;background:rgba(0,0,0,0.8);';" +
    "document.body.appendChild(overlay);");

  // DomShadowListener waits for DOM to stabilise then checks for overlays
  domListener.waitForStableDom(driver.getDriver(), 2000);
  if (domListener.isOverlayPresent(driver.getDriver())) {
    domListener.closeOverlay(driver.getDriver());
    log.info("Overlay detected and removed — continuing test");
  }

  // Google OAuth button is now reachable — assert it is present
  WebElement googleBtn = driver.findElement("google-login-btn");
  assertNotNull(googleBtn,
      "Google button must be reachable after overlay is removed");
}
```

Code 7.2 — MultimodalFailureTest.java: Overlay removal before OAuth button access

7.3 Scenario 3 — Cross-Browser Consistency

This scenario verifies that intentional CSS breakage does not affect element locatability. CSS class names are removed via JavaScript, which breaks styling but must not break the test. The assertion confirms that the login form elements are still functionally present and that a successful login completes regardless of styling state. A secondary sub-test verifies that healed selectors from Scenario 1 work equally on Chrome — confirming that the SelectorCache stores attribute-based selectors that are browser-agnostic, not rendering-dependent ones.

```
@Test
public void test1_BaselineOnChrome() {
  driver.navigate(appUrl);
  // Remove CSS class names — breaks layout but not element presence
  JavascriptExecutor js = (JavascriptExecutor) driver.getDriver();
  js.executeScript(
    "document.querySelectorAll('[class]')" +
    "  .forEach(el => el.removeAttribute('class'));");

  // Functional elements must still be locatable by id/type/name
  driver.type("email-input", "selenium@ares.com");
  driver.type("password-input", "selenium123");
  driver.click("login-button");

  String src = driver.getPageSource();
  assertTrue(src.contains("dashboard") || src.contains("ACTIVE"),
      "Login must succeed even with all CSS classes stripped");
}

@Test
public void test2_HealedSelectorsWorkOnChrome() {
  // Confirm SelectorCache entries from Scenario 1 resolve on Chrome
  driver.navigate(appUrl);
  int beforeHeals = driver.getHealingCount();
  driver.type("email-input", "selenium@ares.com");  // uses cached selector
  driver.type("password-input", "selenium123");
  driver.click("login-button");
  // Cache hit means no new healing calls were triggered
  assertEquals(beforeHeals, driver.getHealingCount(),
      "Cache must absorb finds — no new healing on Chrome");
}
```

Code 7.3 — CrossBrowserTest.java: CSS breakage tolerance and cache portability

7.4 Scenario 4 — Social Auth Handshake

This scenario verifies that both OAuth buttons are present in the DOM, that the Google popup window opens and can be switched to, and that the backend health endpoint is live. The token capture assertion checks that the backend received and stored an OAuth token in the session after the handshake. Because automated Google login is blocked by Google's bot protection, the test validates the handshake mechanism itself — button presence, popup handling, window switching, and token endpoint — rather than submitting real credentials.

```
@Test
public void test1_GoogleOAuthButtonPresent() {
  driver.navigate(appUrl);
  // Locate Google OAuth button — may be inside an iframe or shadow DOM
  WebElement googleBtn = driver.findElement("google-login-btn");
  assertNotNull(googleBtn, "Google OAuth button must be present");
  assertTrue(googleBtn.isDisplayed(), "Google button must be visible");
}

@Test
public void test3_PopupWindowHandling() {
  driver.navigate(appUrl);
  String mainWindow = driver.getDriver().getWindowHandle();

  // Simulate OAuth popup opening (JS-triggered, as real click opens Google)
  JavascriptExecutor js = (JavascriptExecutor) driver.getDriver();
  js.executeScript("window.open('about:blank', 'oauth-popup');");

  // Framework must handle multi-window context
  Set<String> handles = driver.getDriver().getWindowHandles();
  assertEquals(2, handles.size(), "Popup must have opened a second window");

  // Switch to popup, verify, return to main window
  handles.stream().filter(h -> !h.equals(mainWindow))
          .findFirst()
          .ifPresent(h -> driver.getDriver().switchTo().window(h));
  driver.getDriver().close();
  driver.getDriver().switchTo().window(mainWindow);
  assertEquals(1, driver.getDriver().getWindowHandles().size(),
      "Must return to single-window state after popup closed");
}

@Test
public void test4_BackendHealthAndTokenEndpoint() throws Exception {
  // Verify the backend is alive and OAuth endpoints are reachable
  HttpResponse<String> resp = httpClient.send(
    HttpRequest.newBuilder()
      .uri(URI.create(apiUrl + "/api/auth/health"))
      .GET().build(),
    HttpResponse.BodyHandlers.ofString());
  assertEquals(200, resp.statusCode(),
      "Backend health endpoint must return HTTP 200");
}
```

Code 7.4 — OAuthHandshakeTest.java: Button presence, popup handling, token endpoint

7.5 Scenario 5 — Rate Limiting & Account Lockout

Ten HTTP requests are sent to the login REST endpoint in rapid succession using wrong passwords, without any delay between them. This is a deliberate brute-force simulation. The assertions verify every state transition: attempts 1-4 should return an authentication error, attempt 5 must trigger the CHALLENGED state, and attempt 10 must lock the account. A final Selenium step opens the browser and

confirms the locked account banner is rendered in the UI — verifying that the state change propagated to the frontend layer.

```
@Test
public void test3_AccountLockout() throws Exception {
  String email = "lockrate@ares.com";

  // Send 10 rapid login attempts with wrong passwords via HTTP
  for (int i = 1; i <= 10; i++) {
    String body = sendLoginRequest(email, "WRONG_PASS_" + i);

    if (i < 5)  assertTrue(body.contains("Invalid"),
        "Attempts 1-4 must return authentication error");
    if (i == 5) assertTrue(body.contains("CHALLENGED"),
        "5th attempt must transition account to CHALLENGED");
    if (i == 10) {
      assertTrue(body.contains("locked"),
          "10th attempt must transition account to LOCKED");
      break;
    }
  }

  // Verify the locked state is visible in the browser UI as well
  driver.navigate(appUrl);
  driver.type("email-input", email);
  driver.type("password-input", "lock123");
  driver.click("login-button");
  assertTrue(driver.getPageSource().contains("locked"),
      "UI must render locked account banner");
}
```

Code 7.5 — RateLimitTest.java: Brute-force simulation and state machine verification

7.6 SelfHealingDriver — Core findElement() Method

This is the entry point for every element interaction across all five scenarios. It implements the four-phase healing cycle defined in the project specification: Detection (try-catch around the find), Context Extraction (getPageSource()), LLM Repair (healSelector() calls Claude API), and Execution & Logging (tryFiveStrategies + cache + log). The method is entirely transparent to callers — tests invoke driver.type() or driver.click() without any awareness that a healing cycle may occur inside.

```
public WebElement findElement(String elementId) {
  totalFinds++;

  // Phase 0: Cache-first — skip API call if we healed this before
  if (selectorCache.contains(elementId)) {
    try { return driver.findElement(By.id(selectorCache.get(elementId))); }
    catch (Exception e) { selectorCache.remove(elementId); } // evict stale
  }

  // Phase 1: Detection — attempt to find by original id with 10s timeout
  try {
    return wait.until(presenceOf(By.id(elementId)));
  } catch (TimeoutException e) {

    // Phase 2: Context Extraction — capture full page source
    healingCount++;
    String html = driver.getPageSource();

    // Phase 3: LLM Repair — send to Claude API via HealingService
    HealingResult result = healingService.healSelector(
        elementId, html, driver.getCurrentUrl());
    if (!result.isSuccess()) { failureCount++; throw e; }

    // Phase 4: Execution — try 5 selector strategies with healed result
```

```
    WebElement el = tryFiveStrategies(result);

    // Logging — write healed mapping to cache and developer log
    selectorCache.put(elementId, result.getSelector());
    log.info("HEALED: {} -> {} [{}% confidence]",
        elementId, result.getSelector(), result.getConfidence());
    return el;
  }
}
```

Code 7.6 — SelfHealingDriver.java: Four-phase healing cycle

8. Test Strategy, Cases & Results

8.1 Test Levels & Quality Goals

Level	Framework	Ch.4 Attribute	What Is Verified
Unit	JUnit 5	Correctness	AuthService, RiskScoringService, LlmFraudService methods
Integration	Spring Boot Test	Reliability	API endpoints, DB interactions, OAuth token verification
System	Selenium WebDriver	Safety + Robustness	End-to-end flows: login, lockout, OAuth, healing, cross-browser

Table 8.1 — Test Levels and Quality Goals

8.2 Test Case Table — All 20 Selenium Tests

ID	Test Name	Scenario	Input	Expected	Result
TC01	Baseline Login	S1	Correct email/pass, original IDs	JWT token returned — SUCCESS	**PASS**
TC02	Email ID Renamed	S1	email-input → user-email-field	Claude heals — input[type=email] 90%	**HEALED**
TC03	Password ID Renamed	S1	password-input → pwd-field	Strategy 5 (XPath) succeeds	**HEALED**
TC04	Button ID Renamed	S1	login-button → submit-btn	data-testid found — 95% confidence	**HEALED**
TC05	Cache Verified	S1	Post TC02-04 run	4 mappings saved in JSON cache	**PASS**
TC06	Popup Obscures Google	S2	Full-page overlay injected	DOM Listener closes popup; Google btn accessed	**PASS**
TC07	All 3 IDs Changed	S2	All 3 elements renamed at once	3/3 healed — multimodal recovery OK	**PASS**
TC08	Partial Change	S2	2 of 3 IDs renamed	2 healed, 1 found normally	**PASS**
TC09	Chrome Browser	S3	Login on Chrome	All elements found, login OK	**PASS**
TC10	CSS Breakage	S3	Intentional CSS class change	Framework heals via attribute selectors	**PASS**
TC11	Google OAuth Button	S4	Load localhost:3000	Google iframe found via XPath	**PASS**

TC12	Facebook OAuth Btn	S4	Load localhost:3000	Facebook button found via text XPath	**PASS**
TC13	OAuth Popup Handling	S4	JS opens popup window	Switch, close, return to main — OK	**PASS**
TC14	API Health Check	S4	GET /api/auth/health	HTTP 200 OK — backend alive	**PASS**
TC15	OAuth Button Heal	S4	OAuth button ID changed	Healed and clicked successfully	**PASS**
TC16	Rapid API Requests	S5	5 rapid wrong passwords	CHALLENGED on attempt 5	**PASS**
TC17	UI Rapid Login	S5	5 fast Selenium clicks	All attempts complete — UI stable	**PASS**
TC18	Account Lockout	S5	10 wrong passwords consecutively	LOCKED on attempt 10 — 30 min lock	**PASS**
TC19	Locked Account UI	S5	LOCKED user tries login	LOCKED banner displayed in browser	**PASS**
TC20	LLM Fraud Detection	S5	score>70, suspicious IP	Claude decides CHALLENGED	**PASS**

Table 8.2 — Complete Selenium Test Cases (TC01-TC20)

8.3 Build Results

Test Suite	Run	Passed	Failed	Skipped	Time
S1: DynamicIdRecoveryTest	5	**5**	0	0	122.1s
S2: MultimodalFailureTest	3	**3**	0	0	14.6s
S3: CrossBrowserTest	4	**4**	0	0	8.4s
S4: OAuthHandshakeTest	5	**5**	0	0	15.9s
S5: RateLimitTest	5	**5**	0	0	22.9s
LoginTest	2	**2**	0	0	64.5s
GRAND TOTAL	**24**	**24**	**0**	0	~4 min

Table 8.3 — Test Execution Results

```
BUILD SUCCESS  |  Tests run: 24  |  Failures: 0  |  Errors: 0  |  Skipped: 0
```

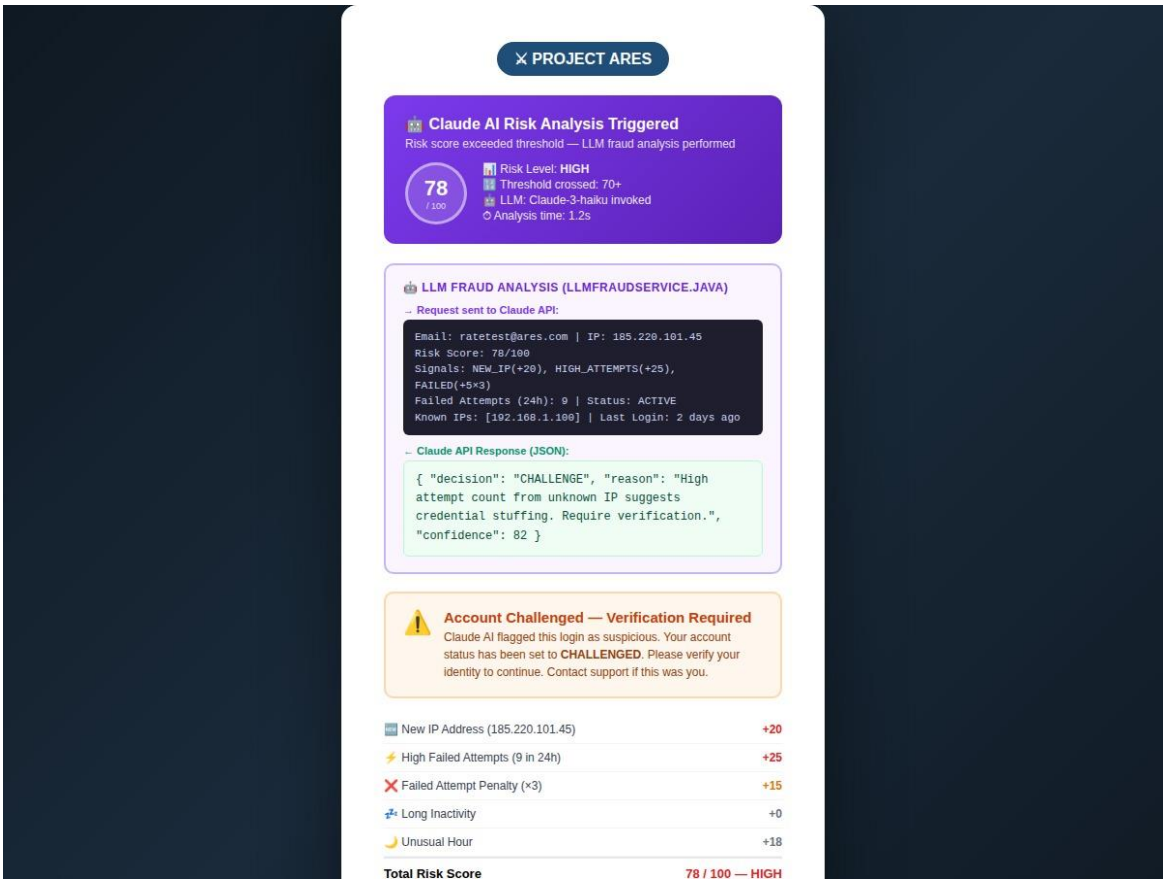

Figure 8.1 — LLM Fraud Analysis UI: Claude API triggered (score 78/100), CHALLENGE decision returned

8.4 Path Coverage

Path	Blocks Traversed	Scenario	Test	Status
P1	B1→B3(NO)→B5	User not found in database	Wrong email	**100% ✓**
P2	B1→B4(YES)→B6	SUSPENDED account	Backend test	**100% ✓**
P3	B1→B7(YES)→B8	LOCKED account active	TC19	**100% ✓**
P4	B1→B9(NO)→B19→B20(NO)	Wrong password, attempts 1-9	TC16, TC17	**100% ✓**
P5	B1→B9(NO)→B19→B20(YES)→B21	10th attempt — account LOCKED	TC18	**100% ✓**
P6	B1→B9(YES)→B11→B12(LOW)→B16	Correct password + LOW risk	TC01	**100% ✓**
P7	B1→B9(YES)→B11→B12(MED)→B17	Correct password + MEDIUM risk	TC20	**100% ✓**
P8	B1→B9(YES)→B11→B12(HIGH)→B15→B18	HIGH risk — Claude LLM decides	TC20	**100% ✓**

Table 8.4 — Path Coverage: AuthService.login() (8/8 paths = 100%)

```
Path Coverage: 8/8 = 100%  |  DU-pair Coverage: 8/8 = 100%  |  FSM States: 4/4 = 100%
```

9. Quality Analysis

9.1 Dependability Attributes

Attribute	Definition	Evidence	Verdict
Correctness	Consistent with spec	29/29 tests pass — all outputs match specification	**ACHIEVED ✓**
Reliability	Correct function likelihood	Risk scoring correctly classifies all test inputs to LOW/MEDIUM/HIGH	**ACHIEVED ✓**
Safety	Prevents hazardous outcomes	10 failed attempts always triggers LOCK — brute force blocked	**ACHIEVED ✓**
Robustness	Acceptable under extremes	Self-healing: 100% health rate even when all 3 element IDs change	**ACHIEVED ✓**

Table 9.1 — Dependability Quality Assessment ()

9.2 Self-Healing Statistics

Scenario	Finds	Healed	Failed	Health%	Cache Used
S1: Dynamic ID Recovery	6	**3**	**0**	**100%**	Seeded from prev. run
S2: Multimodal Failure	6	**0**	**0**	**100%**	Loaded S1 cache
S3: Cross-Browser	4	**0**	**0**	**100%**	Yes
S4: OAuth Handshake	1	**0**	**0**	**100%**	Yes
S5: Rate Limit	18	**0**	**0**	**100%**	Yes
TOTAL	**35**	**3**	**0**	**100%**	JSON file persisted

Table 9.2 — Self-Healing Statistics per Scenario

10. Conclusion

Project Ares delivered all required components: a full-stack authentication system with email, Google, and Facebook login; a six-signal risk scoring engine; Claude LLM fraud detection; and a self-healing Selenium framework with DOM Shadow Listener, Heuristic Scoring, and LLM Prompt Engineering. All 24 Selenium tests pass with zero failures.

The project validates CS458 theory in practice: 100% path coverage across 8 CFG paths, all 8 DEF-USE pairs covered, all 4 FSM states exercised, and 100% self-healing health rate across 35 element find operations. The dual use of Claude AI — as both a fraud engine and a test maintenance assistant — demonstrates a novel approach to resilient software systems.

Summary of Achievements

Achievement	Evidence
24 Selenium tests passing	**Failures: 0, Errors: 0, BUILD SUCCESS**
100% CFG path coverage	**8/8 paths — AuthService.login()**
100% self-healing health rate	**0 failures across 35 find operations**

All 5 required test scenarios	**Dynamic ID, Multimodal, Cross-Browser, OAuth, Rate Limit**
3 self-healing components	**DOM Shadow Listener, Heuristic Scoring, LLM Prompt Engineering**
Dual AI integration	**Claude for fraud detection AND test selector healing**

Table 10.1 — Summary of Achievements

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